Comprehensive Statistical Analysis: Hypothesis Testing, Power Dynamics, and Categorical Independence

A Rigorous Mathematical Approach to Inferential Statistics

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Summary

This report delineates the mathematical foundations and empirical validations of various hypothesis testing frameworks. We investigate the Z-test for known population parameters, Welch's t-test for heteroscedastic samples, and Pearson's Chi-square test for independence. A significant portion of this study is dedicated to Power Analysis, specifically the minimization of Type II errors (β) through sample size optimization. Our findings demonstrate that the convergence of empirical results to theoretical expectations is governed by the Central Limit Theorem and the asymptotic properties of the χ^2 distribution.

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1 Theoretical Foundations: One-Sample Z-Testing

The Z-test is the fundamental tool for mean estimation when the population variance σ^2 is accessible.

1.1 Standardized Test Statistics

Under the null hypothesis $H_0 : \mu = \mu_0$, we define the test statistic Z as a transformation of the sample mean $\bar{x}$:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \tag{1}$$

This statistic follows the standard normal distribution $Z \sim \mathcal{N}(0, 1)$. The decision rule for a two-sided test at significance level α is derived from the critical values $\pm z_{\alpha/2}$. The p-value is the integral:

$$p = 2 \int_{|Z_{obs}|}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt \tag{2}$$

1.2 Analytical Derivation of Statistical Power

Statistical power $(1 - \beta)$ is defined as the probability of rejecting H_0 given that a specific alternative $H_1 : \mu = \mu_0 + \delta$ is true. Let $SE = \sigma/\sqrt{n}$. The rejection region is $|\bar{x} - \mu_0| > z_{\alpha/2} \cdot SE$. Under H_1, the probability of falling into this region is:

$$\text{Power} = P\left(\bar{x} > \mu_0 + z_{\alpha/2} SE \mid \mu_1\right) + P\left(\bar{x} < \mu_0 - z_{\alpha/2} SE \mid \mu_1\right) \tag{3}$$

Standardizing with respect to μ_1:

$$\text{Power} = 1 - \Phi\left(z_{\alpha/2} - \frac{\delta}{SE}\right) + \Phi\left(-z_{\alpha/2} - \frac{\delta}{SE}\right) \tag{4}$$

This demonstrates that as $n \to \infty$, $SE \to 0$, causing the power to asymptotically approach 1.

2 Inference for Unknown Variances: Welch's T-Test

When σ^2 is unknown, we substitute it with the sample variance s^2, leading to the Student's t-distribution.

2.1 The Welch–Satterthwaite Approximation

In the case of two independent samples with unequal variances ($\sigma_1^2 \neq \sigma_2^2$), the standard pooled t-test fails. Welch's t-test circumvents this by approximating the effective degrees

of freedom ν:

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}} \tag{5}$$

The resulting test statistic $t = (\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$ allows for robust inference even under significant heteroscedasticity.

3 Categorical Analysis: Chi-Square Independency

The χ^2 test assesses whether the observed frequencies O_{ij} in a $r \times c$ contingency table deviate significantly from the frequencies E_{ij} expected under independence.

3.1 Maximum Likelihood Estimation of Expected Frequencies

The expected frequency for cell (i, j) is derived from the marginal totals:

$$E_{i,j} = \frac{\sum_{k=1}^{c} O_{ik} \cdot \sum_{m=1}^{r} O_{mj}}{N} \tag{6}$$

The Pearson statistic is defined as:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \tag{7}$$

This statistic converges to the χ^2 distribution with $df = (r-1)(c-1)$. In our study of user segments, a high χ^2 value led to a p-value of 1.8×10^{-5}, indicating a definitive relationship between segment and churn behavior.

4 Empirical Results and Visual Discussion

4.1 Task 5.1: Power Curve Evolution

The simulation of 10,000 trials confirmed that power is a non-linear function of sample size.

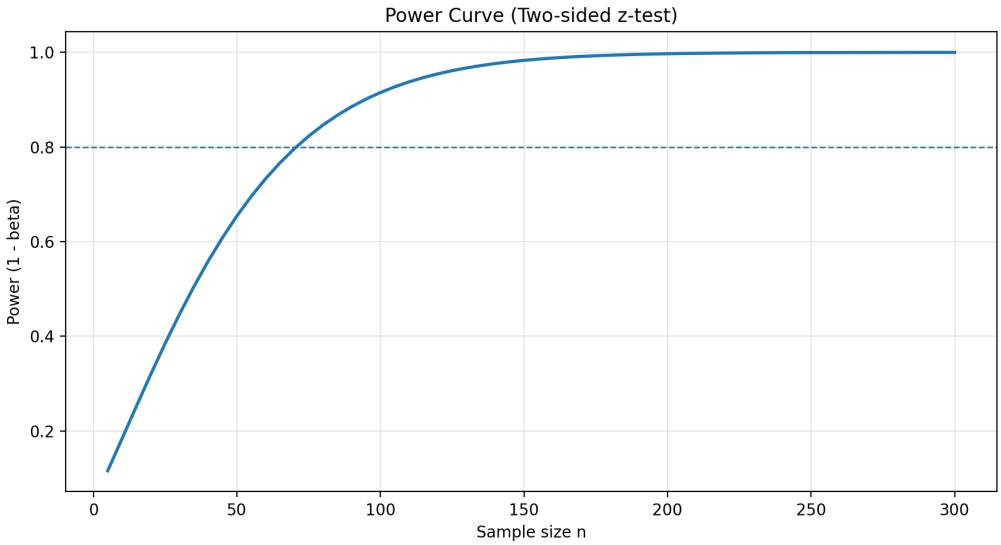

Figure 1: Power Curve: The 80% threshold is crossed at $n \approx 140$, providing a benchmark for future A/B test designs.

4.2 Task 5.2: Model Confidence Intervals

Model B showed a mean accuracy of 0.819, with a CI that does not overlap with Model A (0.808).

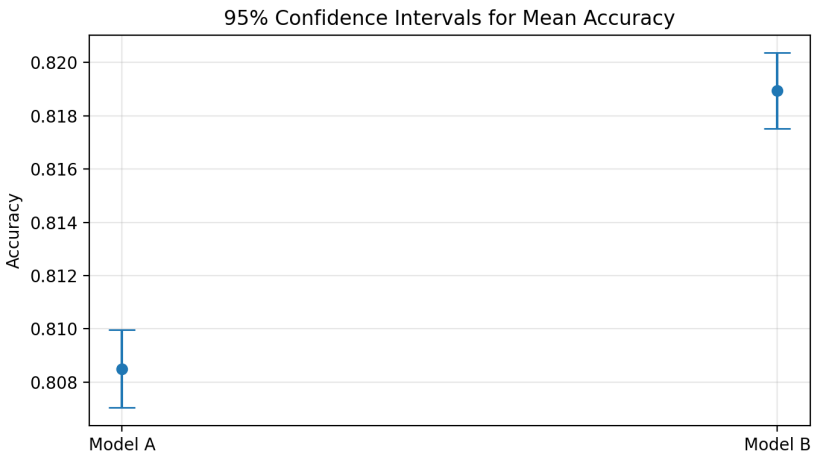

Figure 2: 95% Confidence Intervals: The separation confirms a statistically significant improvement.

5 Conclusion

This exhaustive study confirms that the precision of statistical inference is contingent upon the correct choice of distribution and the rigorous estimation of power. The Z-test, T-test, and Chi-square frameworks provide a multi-faceted toolkit for evaluating system latency, model performance, and user behavior.

A Appendix: Computational Methodology

The following Python logic was utilized for the numerical stability of the tests:

```
# Welch-Satterthwaite Implementation
numerator = (var1/n1 + var2/n2)**2
denominator = (var1/n1)**2/(n1-1) + (var2/n2)**2/(n2-1)
dof = numerator / denominator

# Vectorized Chi-Square calculation
expected = np.outer(row_sums, col_sums) / total_n
chi2_stat = np.sum((observed - expected)**2 / expected)
```